

Deck ROM Sync

Windows Desktop App for Steam Deck Emulation Library Management

User Guide & Reference Manual — v1.1.0

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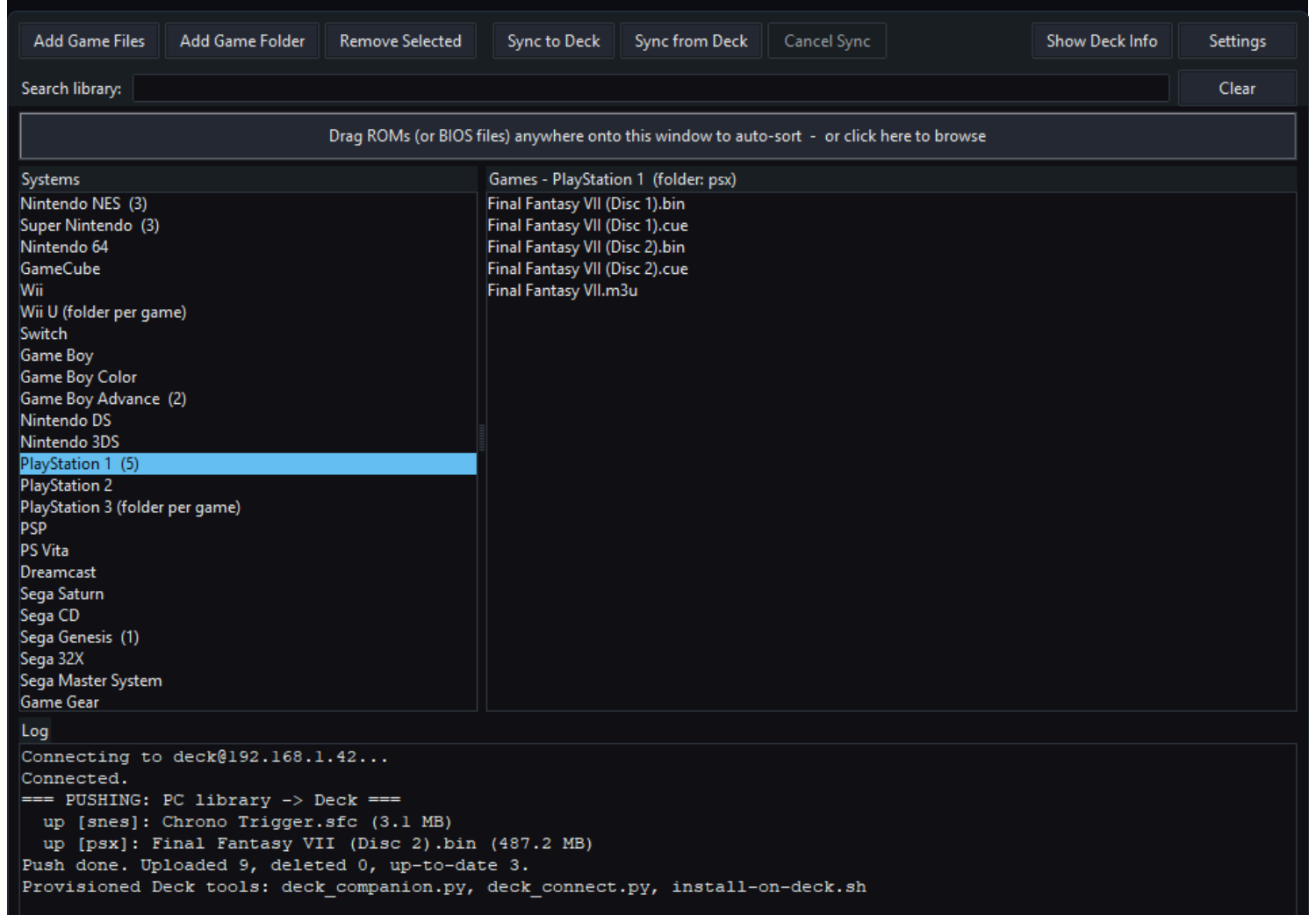
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Deck ROM Sync User Guide

1. Overview

1.1 What Deck ROM Sync Does

Deck ROM Sync is a Windows desktop application that manages your emulator ROM library and mirrors it to a Steam Deck running EmuDeck. You organize your games on your PC and click "Sync to Deck" — the app uploads new games, removes deleted ones, and prepares your ROMs in the exact format EmuDeck expects. Setting up the Deck side is a single double-click; no terminal typing required.



The main window — your library on the left by system, games on the right, sync log at the bottom.

1.2 The Sync Model

Deck ROM Sync uses a push-pull model:

- **Sync to Deck (Push):** Uploads your PC library to the Deck. New or changed files are sent; files deleted from your PC are removed from the Deck, after a confirmation you always see first. This is the main workflow.
- **Sync from Deck (Pull):** Imports ROMs from the Deck into your PC library. Always additive — it never deletes anything already on your PC.
- **The first time you sync** with a given Deck, or whenever your PC library is empty, the app offers to pull from the Deck first, so you never accidentally wipe out existing games.

1.3 Key Features

- **One-click Deck setup:** download a zip, unzip on the Deck, double-click one installer — SSH, ROM/BIOS folders, and the Deck-side helper tools are all configured automatically.
- **Connect Info app:** a Deck-side tool that shows exactly what to type into Settings — no guessing IPs.
- **Whole-window drag-and-drop:** drop ROMs anywhere on the app and they're auto-sorted by system.
- **File preparation:** generates `.cue` files for lone `.bin` images; builds `.m3u` playlists automatically for multi-disc games.
- **BIOS management:** flags required firmware by system and syncs it from your own BIOS pool — never downloaded.
- **Deletion safety:** previews every deletion and asks for confirmation before removing anything from the Deck; a large deletion needs a typed confirmation.
- **Auto-updating:** checks for new versions on launch and can safely replace itself in place.
- **Packaged executable:** a standalone Windows `.exe`, no Python install required to run it.

1.4 Where Everything Lives (Folder Layout)

The download is organized so it's obvious at a glance what to run where:

Folder	What's in it
<code>PC - run this on Windows\</code>	<code>Deck ROM Sync.exe</code> — the only thing a Windows PC needs.
<code>Steam Deck - send this to your Deck\</code>	<code>Deck-ROM-Sync-Deck-Setup.zip</code> — the only thing to send to the Deck.
<code>source\</code>	Python/shell source code, for reference or rebuilding.

`build\`

Build scripts (`build_exe.bat` , the PyInstaller spec, etc.).

This guide (`Deck ROM Sync - User Guide.html` / `.pdf`) and `README.md` sit at the top level alongside `READ ME FIRST.txt` .

2. One-Click Steam Deck Setup

2.1 Installing on the Deck

No Konsole typing required. In **Desktop Mode** on the Steam Deck:

1. Send/copy `Steam Deck - send this to your Deck\Deck-ROM-Sync-Deck-Setup.zip` to the Deck (USB drive, network share, or download it there directly) and unzip it — anywhere is fine.
2. Double-click "**Install Deck ROM Sync.desktop**" inside the unzipped folder.

First time only: KDE will ask to "Trust and Launch" it — that's the standard one-time warning for any downloaded app, not something specific to this installer.

3. The installer creates `~/Emulation/roms` and `~/Emulation/bios` , enables SSH, and installs the Deck-side tools to `~/local/share/deck-rom-sync/` . If this Deck account doesn't have a login password yet (SSH needs one), it offers to open a terminal with `passwd` for you.
4. When it finishes, a "**Deck ROM Sync - Connect Info**" icon appears on the Desktop.

No Steam Deck hardware was available to photograph this step

This guide was written and tested without physical Deck hardware, so the setup screens above are described rather than shown — a fabricated "fake Deck screenshot" would be misleading. What you'll actually see is standard KDE Plasma Desktop Mode: a file manager window, then a one-time "Trust and Launch?" dialog, then a terminal window the installer prints progress to.

After that first install, the PC app re-uploads and refreshes these Deck-side tools automatically every time it connects (see **Show Deck Info**, §6.3) — later updates need zero extra steps on the Deck. Nothing here uses `pip` or touches anything outside your home directory: SteamOS's filesystem is read-only system-wide, so the installer only ever uses the Deck's system Python 3 standard library and tools SteamOS/KDE already ship (`systemctl` , `sudo` / `pkexec` , `passwd` , `kdiallog` / `zenity`).

2.2 The Connect Info App

Double-click the "**Deck ROM Sync - Connect Info**" icon on the Deck's Desktop any time to see everything the PC app needs — copy-what-you-see, no guessing:

- This Deck's IP address(es), labeled Wired or Wi-Fi
- Hostname and username (normally `deck`)
- SSH port (22) and whether `sshd` is currently running — with a one-click "enable it now" offer if it isn't
- The ROMs path (`~/Emulation/roms`) and BIOS path (`~/Emulation/bios`)

It shows this using `kdiallog` (ships with KDE Plasma/SteamOS), falling back to `zenity` or a plain terminal printout if neither is present — either way, no extra install needed. Copy the IP straight into "Steam Deck IP / hostname" in this app's Settings (§3.2).

2.3 Appendix: Manual Setup (Fallback)

If you'd rather not download the installer, the equivalent manual steps in **Desktop Mode > Konsole** are:

```
passwd # set a password for the 'deck' user
sudo systemctl enable --now sshd
```

Find the Deck's IP address via Settings > Internet on the Deck, or `ip a` in Konsole, and use it as "Steam Deck IP / hostname" in Settings.

3. PC Application Setup

3.1 Running the App

Double-click `PC - run this on Windows\Deck ROM Sync.exe` — a single windowed executable, no Python installation required. To rebuild it from source after a code change, see `build\build_exe.bat` in `README.md` .

3.2 Settings Reference

Click **Settings** (top-right) to configure:

Library folder on PC

C:\Users\Tyler\Documents\EmuLibrary

Steam Deck IP / hostname

192.168.1.42

SSH port

22

Deck username

deck

Deck ROMs path

/home/deck/Emulation/roms

Deck BIOS path (blank = roms-sibling default)

Local BIOS pool folder: C:\Users\Tyler\Documents\EmuLibrary\bios

Authentication

☒ SSH key (recommended - see README)

☐ Password

Deck password

(none stored)

☐ Clear stored password

Passwords are encrypted with the Windows Data Protection API (DPAPI), tied to your Windows login on this PC - the file is useless if copied elsewhere or read by another account. Anything already running as YOU on THIS PC can still decrypt it, though, so it's not a substitute for SSH keys, which store no secret on disk at all.

Version 1.1.0

Check for Updates

Save

Settings — library folder, Deck connection details, authentication, and the version/update-check row at the bottom.

Field	What it's for
Library folder on PC	Where your organized ROM collection lives locally.
Steam Deck IP / hostname	Copy this from the Deck's Connect Info app (\$2.2).
SSH port	Default 22 — matches what Connect Info shows.
Deck username	Default <code>deck</code> .

Deck ROMs / BIOS path

Default `/home/deck/Emulation/roms`; BIOS path blank = sibling `bios` folder.

3.3 SSH Key vs. Password Authentication

- **SSH key (recommended):** nothing secret is ever stored on the PC. Generate one in PowerShell:

```
ssh-keygen -t ed25519  
type $env:USERPROFILE\.ssh\id_ed25519.pub | ssh deck@<deck-ip> "mkdir -p ~/.ssh &&  
chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys"
```

Then in Settings, pick **SSH key** and leave the password blank.

- **Password:** encrypted with the **Windows Data Protection API (DPAPI)** before being written to `%APPDATA%\DeckRomSync\config.json`. That means the encrypted blob is useless if copied to another PC or read by a different Windows account — but it *is* still readable by anything running as you on this PC, the same guarantee browsers give saved passwords. SSH keys avoid the question entirely, which is why they're the default.

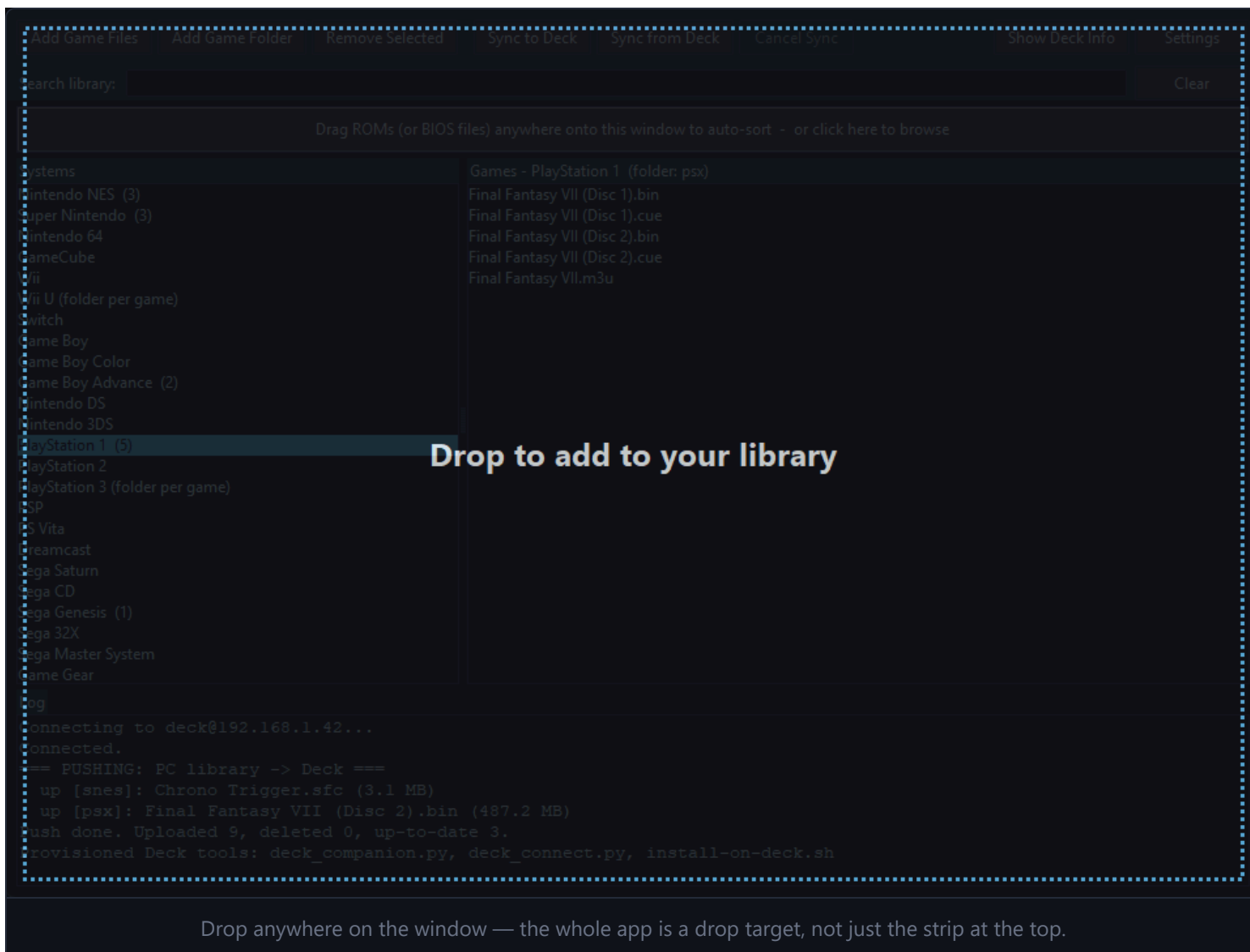
4. Adding Games to Your Library

4.1 Organizing Your ROM Collection

Your PC library mirrors EmuDeck's own folder structure: `<library>\<system>\game files` (e.g. `EmuLibrary\psx\Final Fantasy VII (Disc 1).bin`). You never have to build this by hand — auto-sort does it for you.

4.2 Whole-Window Drag-and-Drop

Drag ROM (or known BIOS) files onto *any part* of the window — not just the drop-zone strip — and a dashed overlay confirms the drop target while auto-sort figures out where each file belongs:



Drop anywhere on the window — the whole app is a drop target, not just the strip at the top.

No drag-and-drop hardware? Use **Add Game Files** / **Add Game Folder** instead — same auto-sort and preparation logic runs either way.

4.3 Auto-Sorting by Extension

Unambiguous extensions (`.nes` , `.sfc` , `.gba` , ...) route straight to their system folder. Ambiguous ones (`.iso` , `.bin` , `.cue` , `.chd` , `.zip` , `.7z`) prompt you to pick the system once.

4.4 Automatic `.cue` / `.m3u` Preparation

- A lone `.bin` gets a minimal `.cue` generated automatically.
- Files named like `Game (Disc 1)` , `Game (Disc 2)` get an `.m3u` playlist built automatically.
- Every `.cue` 's referenced files are checked — a broken reference shows up as a **NEEDS:** warning after import.

5. BIOS Management

5.1 The BIOS Pool Folder

A single `<library>\bios` folder holds firmware you already own. Drop known BIOS files onto the app and they route there automatically (not into a ROM system folder). When a game needs BIOS the pool already has, it's queued to sync to the Deck's `Emulation/bios/` right alongside your games.

5.2 Supported Systems & Requirements

System	Requirement
PlayStation 1	Any one <code>scph550x.bin</code> regional dump
PlayStation 2	Any PS2 BIOS dump (identified by content, not filename)
Sega Saturn	Any one regional dump
Sega CD	Any one regional dump
Dreamcast	<code>dc_boot.bin</code> and <code>dc_flash.bin</code>
Game Boy Advance	<code>gba_bios.bin</code> (optional, checked anyway)

5.3 We Never Download BIOS

BIOS/firmware is never downloaded from anywhere — console BIOS is copyrighted. The app only *flags* what's missing (shown as `NEEDS:` in the log and in a popup after import); you supply firmware from

hardware you own. `deck_companion.py` double-checks presence in `~/Emulation/bios` on the Deck itself after every sync.

6. Synchronization

6.1 Sync to Deck (Push)

Mirrors every system folder to `/home/deck/Emulation/roms/<system>/` over SFTP, with a per-file progress bar and an overall "file n of m " counter.

- **Deletions are always previewed.** If files were removed from your PC library, the app shows exactly what it's about to delete from the Deck and asks once per app session before doing it.
- **Large deletions need a typed confirmation** — if a push would remove at least half of what's currently on the Deck, a plain Yes/No isn't enough; you must type `DELETE`.
- **Pushing from an empty PC library is refused outright** — it would otherwise wipe the Deck to match nothing. You'll be offered a pull instead.
- After a successful push, the app uploads and runs `deck_companion.py` on the Deck, which validates BIOS/ `.cue` / `.m3u` completeness and refreshes the ES-DE game list.

6.2 Sync from Deck (Pull)

Downloads whatever's on the Deck into your matching local folders. **Always additive-only** — pull has no delete path at all, so nothing on your PC can ever be removed this way, mirror mode or otherwise.

6.3 Show Deck Info

Once connected, click **Show Deck Info** in the toolbar to fetch and display the same IP/username/SSH-status/paths the Deck's own Connect Info app shows — a quick, optional confirmation that Settings is pointed at the right place. It changes nothing on the Deck; entirely read-only.

This matches what's currently in Settings - handy confirmation your pairing is correct:

Hostname: steamdeck
Username: deck
IP: 192.168.1.42 (Wi-Fi, wlan0)
SSH port: 22
sshd status: RUNNING
ROMs path: ~/Emulation/roms
BIOS path: ~/Emulation/bios

Close

Show Deck Info — confirms your Settings match what the Deck actually reports.

7. Keeping Deck ROM Sync Up to Date

7.1 How the Auto-Updater Works

A few seconds after launch, the app checks a small version manifest published alongside the download. If a newer version is available, a small, dismissible dialog appears — you can keep using the app while it's open:

Deck ROM Sync 1.2.0 is available (you have v1.1.0).

Adds whole-window drag-and-drop polish and faster BIOS pool scanning.

Later

Update Now

The update-available dialog — non-blocking, shows what's new, "Later" just dismisses it.

- **Later** closes the dialog; it won't nag again until the next launch.
- **Update Now** downloads the new exe, verifies it (size and a Windows-executable signature check, plus an optional checksum), backs up the current exe first, then hands off to a small helper that waits for the app to close, swaps the files, and relaunches automatically.

This is fail-safe by design. Every step before the actual file swap leaves your currently-running app completely untouched if anything goes wrong (no connection, an incomplete download, a write-protected folder) — you'll see a message and the app keeps working normally. A backup copy is also kept next to the exe until the swap is confirmed. A bad update can't brick the app.

The whole check is best-effort and silent by default: no internet, an unreachable host, or a malformed response are all treated as "nothing to report" and never delay or interrupt opening the app.

7.2 Manual "Check for Updates"

Open **Settings** and click **Check for Updates** (bottom of the dialog, next to the current version number) to check on demand — unlike the silent launch check, this always reports back, even if you're already up to date.

8. Troubleshooting & FAQ

Problem	What to do
"paramiko is not installed"	Re-run <code>build\build_exe.bat</code> or <code>build\run.bat</code> , or <code>pip install -r build\requirements.txt</code> manually.
Connection refused / timeout	Double-click "Deck ROM Sync - Connect Info" on the Deck to confirm SSH is running and check the current IP.
Drag-and-drop doesn't work	<code>tkinterdnd2</code> isn't installed; use "Add Game Files" / the drop zone's click-to-browse fallback instead.
Games don't show up on the Deck after sync	The app already runs <code>deck_companion.py --refresh</code> , which restarts ES-DE. If you launch through Steam directly, open EmuDeck > Steam ROM Manager and click "Add games".
Update dialog never appears	Expected if you're already on the latest version, or the update manifest hasn't been republished yet for the latest release — use Settings > Check for Updates to confirm either way.

An update failed partway

By design, your current version stays fully working either way — check the log panel for the specific reason (download, verification, or permissions), fix that, and try again.